Survey 01 — The Landscape of Mechanistic Interpretability

Series note. This is the first survey in `literature/survey/`. It is a *map*: where the 22 archived papers sit, how they group by approach, and — the part written for you specifically — what mathematics they **share** versus where they **fundamentally diverge**. It is deliberately high-altitude; per-paper deep dives and a focused critique of mathematical blind spots are for later surveys (Part VI sketches the agenda).
Audience & conventions. Written for a mathematician + systems engineer who is not an ML specialist. ML jargon is offloaded to the *ML Concepts* primer; the first use of a term defined there is *italicised*. Paper links point into `../papers/`. All 22 papers were read in full for this survey.
Notation. For a linear map $W : A \to B$ we distinguish (per the *ML Concepts* primer's convention) the **dual map** $W^\vee : B^* \to A^*$ — canonical, metric-free (the symbol is non-standard; some write ${}^tW$ or W') — from the **adjoint** $W^* : B \to A$, which exists only once a metric is chosen. W^* always signals an invoked metric; $W^\vee$ never does.

0. The one-paragraph orientation

Mechanistic interpretability (MI) tries to reverse-engineer trained neural networks into human-understandable algorithms. The corpus you collected is almost entirely the **Anthropic / "Transformer Circuits" lineage** (16 of 22 papers) plus a few external landmarks (Redwood's IOI circuit; Nanda et al.'s grokking; Park–Choe–Veitch's linear- representation geometry). Strikingly, this body of work is **mathematically monolithic at its foundation and pluralistic at its methods**: nearly every paper stands on the same four linear-algebraic commitments (Part I), but they split into six methodological camps that operationalise those commitments very differently (Part II). The deepest intellectual action — and the place a mathematician can contribute — is at two seams: (a) the **inner-product / gauge** question that the field mostly leaves implicit (Part III), and (b) the **linear direction vs. nonlinear manifold** fork that has only just opened (Part IV).

Part I — The shared mathematical substrate

Four commitments recur in essentially every paper. They are not independent; they interlock into a single worldview. Understanding these four is most of understanding the field.

I.1 The Linear Representation Hypothesis (LRH): *features are directions*

The foundational modelling assumption: a high-level concept ("this text is in French", "the subject is a person") is encoded as a **direction** d_i (a unit vector, sometimes a ray or cone to fix sign) in some activation space, and the activation vector decomposes as a sparse linear combination

$$x \;\approx\; b + \sum_i f_i(x)\, d_i, \qquad f_i(x) \geq 0,\ \text{few nonzero.}$$

"Linear" means two things at once: features **superpose additively**, and a feature's **intensity** is the scalar coefficient f_i. This single hypothesis underwrites Elhage et al. [3], Bricken et al. [7], Templeton et al. [15], Park et al. [13], and the entire SAE program. It is stated as an explicit, falsifiable hypothesis — and Part IV is about the first serious crack in it.

I.2 Superposition: $\gg d$ *features in* d *dimensions, paid for in interference*

If features are directions and there are far more candidate features than dimensions, the model packs them as an **overcomplete, almost-orthogonal frame**. Two facts of different *character* are in play, and separating them pays off later (Part III):

- *Existence is metric-free.* That you can have more features than dimensions at all is just linear dependence. Writing the **embedding map** $W : e_i \mapsto d_i$ (feature-coordinate space → the width-d activation space), Elhage et al. [3]'s crispest statement is: superposition is present **iff** W **is non-injective** ($\ker W \neq 0$; the directions $\{d_i\}$ linearly dependent) — a kernel-and-rank claim needing no inner product, forced the moment #features $> d$. (The familiar form "W^*W singular" says the same through the adjoint.)
- *The cost is metric-dependent.* What superposition *pays* is **interference**: the off-diagonal of the Gram matrix $G = D^*D$ (built from the adjoint, hence metric-laden). That this stays small — the frame is *almost*-orthogonal — is a claim about an inner product, underwritten by *Johnson–Lindenstrauss* (exponentially many near-orthogonal unit vectors exist in $\mathbb{R}^d$) and *compressed sensing* (sparse codes are recoverable from low-dimensional linear maps). *Which* inner product is left implicit — exactly the question Part III says the field rarely confronts.

Everything downstream is a negotiation with that Gram matrix.

I.3 The transformer is *almost* linear: residual stream, path expansion, QK/OV

Elhage et al. [1] is the keystone. Its move: the *residual stream* is a purely **linear communication channel** (blocks add their outputs), so the forward pass can be **path-expanded** into a sum of end-to-end linear paths, with the *softmax* attention pattern A as the only nonlinearity. **Freeze** A and the model is an exact linear function of the input tokens. Two low-rank objects fall out of every attention head:

$W_{QK} : V \to V^*$ (a bilinear form — *where* to attend; coordinates $W_Q^\top W_K$), $\qquad W_{OV} = W_O W_V : V \to V$ (an end

The type signature *is* the where/what split: that W_{QK} lands in the **dual** V^* (a pairing of two stream vectors into a score) is exactly why a transpose appears in its coordinate form $W_Q^\top W_K$, while W_{OV} — a map of stream vectors to stream vectors — needs none; see the *ML Concepts* primer §1.4 and the adjoint/dual convention there. "*Virtual weights*" $W_{\text{in}}^{(2)} W_{\text{out}}^{(1)}$ connect any two blocks through the stream. This algebra — tensor/Kronecker products, low-rank factorisations, eigenanalysis of the OV "copying" matrix — is the literal skeleton of Groups 1 and 4 below, and reappears as the QK bilinear form in Group 5.

I.4 No privileged basis (a $GL(V)$ gauge symmetry) — except where a nonlinearity breaks it

Because the residual stream is read and written only by linear maps, you can conjugate the whole network by any invertible $M \in GL(V)$ — replace each read map W_{in} by $W_{\text{in}}M^{-1}$ and each write map W_{out} by MW_{out} — and get a **bit-identical function**. The residual stream therefore has **no privileged basis**: its coordinates are individually meaningless. (Two things *do* survive this gauge: a read map's **kernel** and a write map's **image** transform covariantly, $\ker \mapsto M \ker$, $\operatorname{im} \mapsto M \operatorname{im}$, so *containments between them* — and hence inter-block **virtual weights** $W_{\text{in}}^B W_{\text{out}}^A$ — are invariant, whereas any complement, norm, or angle is not; see the *ML Concepts* primer §1.3.) A *pointwise nonlinearity* (ReLU on MLP neurons) breaks the symmetry and singles out a **privileged basis** — which is why neuron-level interpretability is even conceivable. Olah [4] names this; Elhage et al. [8] shows the symmetry is *empirically* broken in the residual stream too (outlier coordinates, kurtosis $\gg 3$) and fingers the *Adam* optimiser's per-coordinate normalisation as the culprit.

> **Why these four are really one idea.** I.1 says meaning lives in directions; I.4 says the ambient space has no canonical directions; I.2 says we cram more directions than dimensions; I.3 says the dynamics shuttling those directions around are (frozen-)linear. Put together: *the model computes by moving an overcomplete frame of direction-encoded features through a gauge-symmetric linear channel punctuated by sparse nonlinearities.* Hold that sentence; the whole field is footnotes to it.

Part II — The six approaches

#	Approach (camp)	Core question	Papers	Signature mathematics
1	**Algebraic circuit analysis**	Rewrite the forward pass into readable terms	Elhage et al. [1], Olsson et al. [5], Wang et al. [6]	tensor/Kronecker products, low-rank W_{QK}, W_{OV}, path expansion, OV eigenvalues, causal patching
2	**Superposition geometry**	How do $\gg d$ features pack into d dims?	Elhage et al. [3], Henighan et al. [9], Olah et al. [11], Elhage et al. [8], Anthropic Interpretability Team [17]	Thomson problem, uniform polytopes / tegum products, JL, compressed sensing, Gram matrix D^*D, phase transitions, kurtosis test
3	**Dictionary learning (SAEs)**	*Recover* the feature frame from activations	Bricken et al. [7], Templeton et al. [15], Elhage et al. [2], Lindsey et al. [14]	overcomplete dictionary learning, ℓ_1/ℓ_0 sparse coding, encoder/decoder frames, scaling laws
4	**Circuit tracing / attribution**	Read off causal computation graphs on a real model	Ameisen, Lindsey, et al. [16], Lindsey et al. [22], Kamath et al. [21], Jermyn et al. [20]	local linearization (Jacobian/Taylor), stop-gradients, Neumann series $(I - A)^{-1}$, transcoders, interference-corrected virtual weights
5	**Representation geometry**	Take the *geometry* (metric, curvature, group structure) seriously	Park et al. [13], Gurnee et al. [19], Nanda et al. [10]	non-Euclidean inner products, Riesz duality, differential geometry of curves, Fourier analysis / characters of $\mathbb{Z}/n\mathbb{Z}$
6	**Epistemics / vision**	What makes an interpretation *correct*? Where is this going?	Olah [12], Anthropic Interpretability Team [18], Olah [4]	faithfulness vs. function-equality, Jacobian matching, the program's axioms

Group 1 — Algebraic circuit analysis

The exact-rewrite tradition. Elhage et al. [1] treats the (attention-only) transformer as a tensor-algebraic object and multiplies it out; 0-layer models are bigrams $W_U W_E$, 1-layer models are skip-trigrams readable from $W_U W_{OV} W_E$ and $W_E^\vee W_{QK} W_E$ (the dual map $W_E^\vee$ pulls the QK form back to token space), and 2-layer models gain power through **composition** of heads, yielding the

induction head. Olsson et al. [5] elevates that single circuit to a theory of *in-context learning* and ties its sudden formation to a **phase change** in training. Wang et al. [6] (the one Redwood paper) reverse-engineers a 26-head circuit in GPT-2 and contributes the field's first **formal validation criteria** — *faithfulness, completeness, minimality* — plus **path patching**. Mathematically this group is the most classical: it is linear algebra and causal-intervention bookkeeping. Its open wound, stated in Elhage et al. [1] itself, is that **MLPs resist the rewrite** (the nonlinearity won't freeze) — which is the cue for Group 3.

Group 2 — Superposition geometry

The forward-geometry tradition: given features and sparsity, what packing does training choose? Elhage et al. [3] is the gem here: in controlled ReLU autoencoders it shows superposition is governed by a sparsity-vs-importance **phase transition**, and that the learned feature directions self-organise into **solutions of a generalized Thomson problem** — antipodal pairs, pentagons, tetrahedra, square antiprisms — often decomposing as **tegum (orthogonal) products** of low-dimensional uniform polytopes. The capacity question gets a compressed-sensing answer: #features is linear in d but modulated by sparsity. The companion notes specialise it: Henighan et al. [9] (finite data → memorised *datapoints* live in superposition as polygon vertices; double descent falls out); Olah et al. [11] (separates **composition** from **superposition** as two ways to spend the same $\exp(d)$ representational volume — and argues they are *in tension*); Elhage et al. [8] (the symmetry-breaking diagnostic); and Anthropic Interpretability Team [17] (the *weights* between superposed features are themselves forced into superposition, producing spurious "interference weights" = noise). This group is where the most beautiful, most classical mathematics already lives.

Group 3 — Dictionary learning (Sparse Autoencoders)

The inverse problem to Group 2. If the model stores an overcomplete sparse code, **recover the dictionary**. Bricken et al. [7] trains *SAEs* on a one-layer model's MLP and extracts thousands of monosemantic features; Templeton et al. [15] scales this to Claude 3 Sonnet (millions of features, scaling laws for SAE training, a sigmoid law relating a concept's training-frequency to whether it gets a dedicated feature). Elhage et al. [2] is the contrarian: instead of *recovering* features post-hoc, *change the architecture* ($\mathrm{SoLU}(x) = x \odot \mathrm{softmax}(x)$) so features align to neurons in the first place — but a needed LayerNorm "smuggles superposition back", which the authors read as evidence *for* the superposition hypothesis. Lindsey et al. [14] generalises SAEs to **crosscoders** (one dictionary across many layers/models). The whole group is **dictionary learning / sparse coding** with a learned overcomplete dictionary and an ℓ_1 surrogate for ℓ_0; the recurring anxiety is that there is **no ground-truth objective** — reconstruction-MSE-plus-sparsity is only a *proxy* for interpretability.

Group 4 — Circuit tracing / attribution

Scale Group 1's exact rewrite to real MLP-bearing models by replacing "exact" with "locally linear". Ameisen, Lindsey, et al. [16] swaps each MLP for a **cross-layer transcoder** (a Group-3-style sparse feature model that predicts *computation*, not just representation), freezes attention and norm denominators, adds per-token **error nodes** to match the model exactly on one prompt, and reads off a **linear attribution graph**. The math is *local linearization*: stop-gradients make each feature's pre-activation an exact linear sum of upstream contributions (a Taylor expansion, exact at the prompt, divergent away from it); node influence is the **Neumann series** $(I - A)^{-1} - I$. Lindsey et al. [22] applies it to Claude 3.5 Haiku across $\sim$10 case studies (planning, multi-hop reasoning, multilingual circuits, refusals). Kamath et al. [21] and Jermyn et al. [20] attack the part the method *freezes* — the QK bilinear form $V \to V^*$ — by decomposing the attention score into **feature–feature pairings** $\langle W_{QK} x_k, x_q \rangle$. This group is methodologically dominant today and is where "*mechanistic faithfulness*" becomes the central worry (Group 6).

Group 5 — Representation geometry

The mathematically deepest group, and the one most aligned with your interests. It treats the geometry as geometry rather than as bookkeeping. Park et al. [13] formalises "feature = direction" via counterfactual concept pairs, distinguishes the **embedding** (Λ, input/context side) and **unembedding** ($\bar{\Gamma}$, output/word side) spaces — which are *canonically dual*, $\Lambda \cong \bar{\Gamma}^*$, the softmax pairing $\langle \lambda, \gamma \rangle$ needing no metric — and — crucially — observes that the Euclidean inner product is **not canonical** (see Part III), introducing the **causal inner product** — the metric $g_C = \mathrm{Cov}(\gamma)^{-1} : \bar{\Gamma} \xrightarrow{\sim} \bar{\Gamma}^*$, so $\langle \bar{\gamma}, \bar{\gamma}' \rangle_C = \langle g_C\, \bar{\gamma}, \bar{\gamma}' \rangle$ — and a **Riesz-duality** theorem unifying the two spaces. Gurnee et al. [19] finds that a scalar *count* is represented not as a direction but as a **curved 1-D manifold** (a rippling helix in a $\sim$6-D subspace), and that an attention head **compares** two counts by a W_{QK} that **rotates/twists one manifold onto another** — differential geometry, not just linear algebra. Nanda et al. [10] reverse-engineers modular addition as **Fourier multiplication**: inputs are placed on circles $(\cos \omega_k a, \sin \omega_k a)$ (the **characters of** $\mathbb{Z}/n\mathbb{Z}$) and addition becomes angle-addition, read out by trig identities — representation theory of a finite abelian group. Note the resonance: **circular / periodic structure appears independently** in grokking (group characters) and in counting (the "ringing"/Gibbs geometry of the count manifold). That is a genuine shared mathematical phenomenon hiding under two different tasks.

Group 6 — Epistemics / vision

What does it mean to be right? Olah [4] sets the program's vocabulary (network $\approx$ compiled binary; features $\approx$ variables). Olah [12] is the manifesto: once superposition is solved, "for small enough circuits, statements about their behaviour become questions of *mathematical reasoning*" — an explicitly **axiomatic** ambition. Anthropic Interpretability Team [18] is the cold shower: a transcoder can be monosemantic, sparse, and low-error **yet implement a different algorithm** that generalises differently off-distribution. Its proposed fix — **Jacobian matching** (penalise $\|J_{\mathrm{MLP}} - J_{\mathrm{transcoder}}\|_F^2$) — is mathematically the most pointed idea in the group, and it quietly concedes that matching input–output behaviour (Group 3's objective) is *not* matching mechanism.

Part III — What the groups SHARE (the connective tissue)

This is the section you asked to be emphasised. Under the methodological diversity, the groups are bound by a small number of mathematical objects. Seeing the corpus this way is, I think, the single most useful reframe.

III.1 Almost everything is a bilinear pairing — of two kinds: metric-free vs. metric-dependent

Track the inner products through the corpus and nearly every quantity is a **bilinear pairing of two vectors**. The distinction that matters — and that the field rarely flags — is *which kind*, because the two behave oppositely under the $GL(V)$ gauge of I.4.

- **Metric-free (gauge-invariant) pairings** are built only from the model's own maps, or from the natural contraction of a space with its dual; they need no choice of inner product:
  - the **canonical pairing** $\langle \lambda(x), \gamma(y) \rangle$ the *softmax* consumes (III.2) — embedding contracted with unembedding, legitimate because the embedding space is the dual of the unembedding space, $\Lambda \cong \bar{\Gamma}^*$ (no metric needed);
  - the **attention score** $\langle W_{QK} x_k,\, x_q \rangle$ — a *learned* bilinear form $W_{QK} : V \to V^*$, invariant because its query- and key-sides transform contragrediently (Kamath et al. [21] expands it into feature–feature terms);
  - a **linear probe** $\langle w, x \rangle$ — a learned covector $w \in V^*$ evaluated on a vector.
- **Metric-dependent pairings** pair two vectors *of the same space through the standard inner product* — equivalently they apply the **adjoint** (W^*, not the dual $W^\vee$; see the *ML Concepts* primer's convention), silently inserting $G = I$ as a metric the architecture never supplied:
  - **interference / coherence** — the off-diagonal of the feature Gram matrix $G = D^*D$ (Group 2); capacity, adversarial vulnerability, and the Thomson geometry are read off G;
  - **SAE reconstruction** $\|x - \hat{x}\|_2^2$ and the near-orthogonality $G \approx I$ that makes recovery work (Group 3); *feature splitting* is G revealing finer block structure;
  - **cosine similarity** between two feature directions (the default tool for clustering and comparing features).

The operational core of Groups 2–4 lives in the *metric-dependent* column — yet I.4 says the residual stream has no canonical metric. The one paper that treats the metric as something to **derive** rather than default to I is the LRH paper, whose **causal inner product** answers "which G?" with $\mathrm{Cov}(\gamma)^{-1}$ (a whitening / Mahalanobis form). So:

> *Mathematically, the field is the study of one metric-free pairing (embedding·unembedding, plus the learned form W_{QK}) alongside a host of metric-dependent Gram computations that quietly assume the very Euclidean structure the architecture declares arbitrary.*

III.2 The gauge symmetry, and the quantity that actually survives it

Combine I.4 (no privileged basis) with the LRH paper's identifiability law and you get a genuinely important — and, in the corpus, almost entirely **unexploited** — observation.

The softmax output depends only on the canonical pairing $\langle \lambda(x), \gamma(y) \rangle$, and is invariant under the simultaneous change (LRH paper, eq. 3.1)

$$\gamma(y) \;\mapsto\; A\,\gamma(y) + \beta, \qquad \lambda(x) \;\mapsto\; (A^\vee)^{-1}\lambda(x), \qquad A \in GL(\bar{\Gamma})$$

(in coordinates $(A^\vee)^{-1} = A^{-\top}$). This is the **readout instance of the I.4 gauge** (A acting on the final-layer spaces): the unembedding side transforms by A, the embedding side by the inverse **dual**

map $(A^\vee)^{-1}$ — *contragrediently* — so the pairing $\langle \lambda, \gamma \rangle$ is invariant (the additive β merely shifts every logit by the same y-independent amount, which softmax discards — §1.2). That λ transforms as a covector is just the statement $\Lambda \cong \bar{\Gamma}^*$. The general principle, across every activation space:

> **Gauge-invariant means built only from the model's own maps and the natural dual pairing.** Quantities assembled from learned weights and the contraction of a space with its dual — the embedding–unembedding pairing $\langle \lambda, \gamma \rangle$, an attention score $\langle W_{QK} x_k, x_q \rangle$, a probe $\langle w, x \rangle$ — are invariant. Quantities that pair two vectors of the **same** space through the **standard** inner product — the cosine of two feature directions, $\|x - \hat{x}\|_2^2$, the Gram off-diagonal — secretly insert the identity as a metric, and so are **gauge-dependent**: not canonically meaningful without first *choosing* that metric.

This has a sharp, slightly uncomfortable consequence for Group 3. An SAE minimises $\|x - \hat{x}\|_2^2$ in the residual stream and compares decoder directions by Euclidean cosine similarity — both gauge-dependent quantities — while Group 2/4 insist the same space has *no privileged basis.* The practical escape hatch is empirical: training + Adam *do* break the symmetry (Elhage et al. [8]), so the data-induced metric isn't arbitrary. But that is an empirical crutch, not a principled choice of metric — and exactly one paper (the LRH paper) treats the metric as something to be *derived* rather than assumed. **This is the central seam of the whole corpus** and I flag it as the leading candidate for mathematical contribution (Part VI).

The constructive flip side is worth stating, because the gauge problem is not a dead end: there *is* a gauge-invariant way to ask whether two components interact — not the cosine of their directions, but whether one's **image lands in the other's kernel**, equivalently whether their **virtual weight** $W_{\text{in}}^B W_{\text{out}}^A$ vanishes (I.4, the *ML Concepts* primer §1.3). Re-expressing interference and feature interaction in those metric-free terms — rather than through the $G = I$ Gram matrix — is one concrete shape the contribution in Part VI(1) could take.

III.3 The same low-rank attention algebra, reused five times

$W_{QK} : V \to V^*$ (the bilinear form; coordinates $W_Q^\top W_K$) and $W_{OV} = W_O W_V : V \to V$ (the endomorphism) are introduced in Elhage et al. [1] and then *never leave*: induction-head prefix-matching is a positive-eigenvalue OV plus a K-composition QK; the IOI circuit is classified entirely in these terms; circuit tracing freezes the QK and linearises the OV; the attention papers decompose QK into features; the manifold paper finds QK implementing a rotation. **If you learn one piece of transformer algebra, learn the QK/OV factorisation** — it is the corpus's universal coordinate system.

III.4 Three regularizers, one Lagrangian shape

Groups 2–4 and 6 all minimise *reconstruction + sparsity*: $\mathbb{E}\|x - \hat{x}\|^2 + \lambda\, \rho(f)$, with ρ an ℓ_1 surrogate for ℓ_0. Toy models, SAEs, transcoders, crosscoders, the interference-weight model, the unfaithfulness model — same objective, different x, different decoder constraints. The shared pathologies (shrinkage from ℓ_1, dead features, the ℓ_0/ℓ_1 gap exploited by "gradient masking") are therefore *shared* across the whole tooling stack, not quirks of one method.

III.5 Compressed sensing as the two-way bridge

Group 2 uses compressed sensing to *upper-bound* how much superposition is possible (a forward statement about W); Group 3 *is* the recovery algorithm compressed sensing promises (an inverse statement about recovering f from $x = Wf$). Same theorem (RIP / incoherence / sparse recovery), read in both directions. This is the cleanest example of two camps sharing one piece of mathematics.

Part IV — Where the math FUNDAMENTALLY diverges

Genuine forks, not just different tools. These are the fault lines worth your attention.

IV.1 Feature-as-direction (linear) vs. feature-as-manifold (nonlinear) — the big one

Groups 1–4 assume a feature is a **1-D ray**: meaning is a direction, intensity is a scalar. Gurnee et al. [19] breaks this for *continuous* quantities: a count is a **curved 1-dimensional manifold** with deliberately high curvature ("ringing"), embedded in ∼6 dimensions, and the model exploits the curvature (a ray could only be scaled/compared by a monotone product, whereas a curve admits a genuine **rotation** that implements thresholded comparison). This is a *category* difference: linear subspace vs. nonlinear submanifold; linear algebra vs. **differential geometry**. The question it raises is pointed and, as far as the corpus shows, open: **is the Linear Representation Hypothesis simply false for continuous/ordered variables, and right only for discrete categorical ones?** If so, the field needs a geometry of feature *manifolds* (intrinsic dimension, curvature, how attention acts as a near-isometry between them) that currently does not exist in worked-out form. The LRH paper and the manifold paper are, in a precise sense, **mathematically incompatible accounts of what a feature is** — and nobody has reconciled them. (Note the entanglement with III: the manifold's "curvature" and "ringing" are themselves read through the ambient inner product — cosine-similarity matrices of activations — so a properly metric-free account of feature *manifolds* is doubly open.)

IV.2 Modelling representation vs. modelling computation

SAEs (Group 3) model the **representation** — a basis for activation *space*. Transcoders and attribution graphs (Group 4) model the **computation** — the map between activations. The Anthropic Interpretability Team [18] paper shows these are **not interchangeable**: an SAE "can't really mislead you" (it just describes a vector), but a transcoder that reproduces the input–output map can still implement the *wrong algorithm*. The mathematical content of the distinction: representation-modelling constrains a single space; computation-modelling must match a **Jacobian** (a linear map between spaces), and matching a function is strictly weaker than matching its derivative everywhere. This is a real divergence in what "correct" even means, and it cuts the toolkit in half.

IV.3 Exact algebra vs. local linearization vs. learned approximation

Three incompatible notions of "linear" coexist (see the *ML Concepts* primer §1.4 and §3):

1. **Exact, by freezing the nonlinearity** (Elhage et al. [1]): attention frozen ⇒ the forward pass is *honestly* linear in tokens. No approximation.

2. **Local linearization** (Ameisen, Lindsey, et al. [16], attribution patching): a Taylor/Jacobian expansion, *exact at one prompt, wrong away from it.*
3. **Learned sparse approximation** (SAEs/transcoders): a trained model that is neither exact nor a derivative, only a low-error fit — and possibly mechanistically unfaithful.

These are routinely all called "linear", but they have different error semantics, and conflating them is where overclaiming creeps in. A mathematician will want to keep them strictly separate.

IV.4 Discrete-algebraic vs. continuous-geometric task structure

Nanda et al. [10] finds **finite group representation theory** ($\mathbb{Z}/n\mathbb{Z}$ characters); the manifold paper finds **continuous differential geometry**; the superposition papers find **discrete polytopes** (Thomson). These are three different branches of mathematics describing "the geometry of representations", and the field has **not** unified them. The tantalising hint (III, and the grokking↔counting circle resonance) is that **harmonic analysis / periodicity** might be the common generalisation — Fourier on a group, Fourier features on a manifold, and the eigen- structure of a circulant Gram matrix are not unrelated — but this is conjecture, not established.

Part V — A few honest observations about the corpus

- **Provenance bias.** 16/22 papers are one lab's (Anthropic) thread; they share authors, assumptions, and a house style (many are explicitly "preliminary lab-meeting notes"). The shared substrate of Part I is partly a sociological fact, not only a mathematical necessity. The three external papers (IOI, grokking, LRH-geometry) are also the three that contribute the most *formal* machinery (validation metrics; Fourier reverse-engineering; the metric question) — worth noting.
- **Rigour gradient.** Formality decreases as models scale: toy models get closed-form loss analysis and named polytopes; frontier-model papers get qualitative case studies with ∼25% success rates and heavy hedging. The mathematically tractable results are mostly in Group 2 (toy models); the high-stakes claims are mostly in Group 4 (frontier circuits), where faithfulness is weakest.
- **Proxy objectives everywhere.** No one has a ground-truth definition of "feature" or a principled SAE objective; MSE+ℓ_1 is admitted to be a stand-in. Several "definitions" (copying matrix via positive eigenvalues; interference weights; faithfulness) are explicitly heuristic and acknowledged as not-yet-formalisable.

Part VI — Agenda for later surveys (where the math might be)

Signposts, not conclusions. Each is a candidate "mathematical blind spot → contribution" to develop in a focused follow-up.

1. **A gauge theory of representations (III.2).** Make the $GL(V)$ symmetry explicit; characterise the invariants (the embedding–unembedding pairing, learned forms like W_{QK}, and image-in-kernel / virtual-weight relations); re-express superposition/SAE/probing results in gauge-invariant form; ask what an SAE *should* minimise if not the gauge-dependent ℓ_2 loss. The LRH paper's $\mathrm{Cov}(\gamma)^{-1}$ is one answer for the output space; the residual-stream interior is wide open.

2. **A geometry of feature manifolds (IV.1).** Reconcile direction-features with manifold- features. What is the right invariant description of a count/colour/date manifold (intrinsic dimension, curvature, the action of W_{QK} as a near-isometry)? Is LRH a categorical- variable special case of a manifold theory?
3. **The interference Gram matrix as the central object (III.1).** A unified treatment of $G = D^*D$ — spectrum, block structure (feature splitting), the relation between its off-diagonal (interference) and downstream "interference weights", and bounds linking sparsity, dimension, and the operator norm of $G - I$.
4. **Error semantics of the three "linearities" (IV.3).** Make precise when local linearization / learned approximation incurs how much error, and connect Jacobian-matching (mechanistic faithfulness) to a quantitative bound.
5. **Harmonic analysis as a possible unifier (IV.4).** Is there a single framework (Fourier on groups / spectral geometry / circulant structure) underlying the polytope, character, and manifold geometries? Likely the most speculative, possibly the most fruitful.

Appendix — paper → group quick index

- **framework** — G1 (substrate for all)
- **induction-heads** — G1
- **interpretability-in-the-wild-ioi** — G1 (+ validation metrics)
- **toy-models-superposition** — G2 (substrate for G3)
- **superposition-memorization-double-descent** — G2
- **distributed-representations-composition-superposition** — G2
- **privileged-bases-residual-stream** — G2 (substrate I.4)
- **toy-model-of-interference-weights** — G2 (bridges to G4)
- **towards-monosemanticity** — G3
- **scaling-monosemanticity** — G3
- **softmax-linear-units** — G3 (architectural variant)
- **sparse-crosscoders-model-diffing** — G3 (bridges to G4)
- **circuit-tracing-computational-graphs** — G4 (method)
- **on-the-biology-of-a-large-language-model** — G4 (application)
- **tracing-attention-feature-interactions** — G4/G1 (QK)
- **progress-on-attention** — G4/G1 (QK)
- **linear-representation-hypothesis-geometry** — G5 (the metric question)
- **when-models-manipulate-manifolds-counting** — G5 (the manifold fork)
- **progress-measures-for-grokking** — G5 (group/Fourier)
- **interpretability-dreams** — G6
- **mech-interp-variables-bases** — G6 (substrate I.4)
- **toy-model-of-mechanistic-unfaithfulness** — G6 (faithfulness)

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